

Benjamin Chung

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EDUCATION

University of British Columbia

Expected Grad: May 2028

Bachelor of Applied Science – Electrical and Computer Engineering

EXPERIENCE

Junior Electrical Designer

September 2025 – May 2026

Smith + Andersen

- Designed SLDs, power, and lighting layouts for tenant buildings using AutoCAD MEP, Revit, and Visual Lighting.
- Quantified existing electrical equipment and sized feeders, circuit breakers, mech equipment, and panels according to BCBC and CEC.

Airside Project Management

January 2025 – August 2025

Vancouver Airport Authority (YVR)

- Supported YVR's North Runway Program by assisting with technical coordination, project documentation, and stakeholder communication across paving, drainage, and electrical work scopes.
- Created a manual to familiarize new users with YVR's Airfield Lighting System and Power Distribution System with adherence to TP312 aerodrome compliance standards.

PROJECTS

Self-Balancing Robot | *C++, Python, Control Systems, Electronics, Firmware*

- Implemented a cascaded PID architecture to calculate target angle in real-time.
- Fine-tuned gyro and accelerometer filter values, PID gains, and encoder & motor response characteristics.
- Integrated BLE connection with a Flutter mobile app to control robot movements.

iPod Machine with Volume Indicator | *SystemVerilog, SignalTap, FPGA, Firmware*

- Designed a keyboard-input FSM to control flash memory writes.
- Designed a separate FSM to synchronize address controller and flash memory reads.
- Programmed an ISR to sample, visualize, and real-time filter audio data from memory using an embedded microcontroller.

Chat Room | *Python, RTOS, TCP/IP, JSON, Multithreading*

- Built a multi-client and server-hosting TCP chat application using Python sockets.
- Implemented multithreading, socket disconnect detection, messaging/status broadcasts using JSON packet serialization.
- Used mutex lock to preserve asynchronous message orders and prevent race conditions.

RC4 Brute-Force Message Decryptor | *SystemVerilog, SignalTap, FPGA*

- Designed a 16-state FSM decryptor with pipelined reads and a start-finish controller communication protocol with integrated RAM and ROM modules.
- Parallelized brute-force key search across four processing cores to accelerate search throughput.

TECHNICAL SKILLS

Programming/HDL Languages: Python, C/C++, Java, SystemVerilog, Assembly

Proficient Programs: LTSpice, Matlab, Visual Studio, Git, AutoCAD, Revit, Quartus, SignalTap, MS365, Bluebeam

Awards: UBC Centennial Entrance Award, BC Achievement Scholarship

Hobbies: Hiking, Volleyball